

Spatio-Temporal Forecasting System for Hourly Weather Data

Integrating Database, Kriging, and Machine Learning

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Spatial Statistics Final Project

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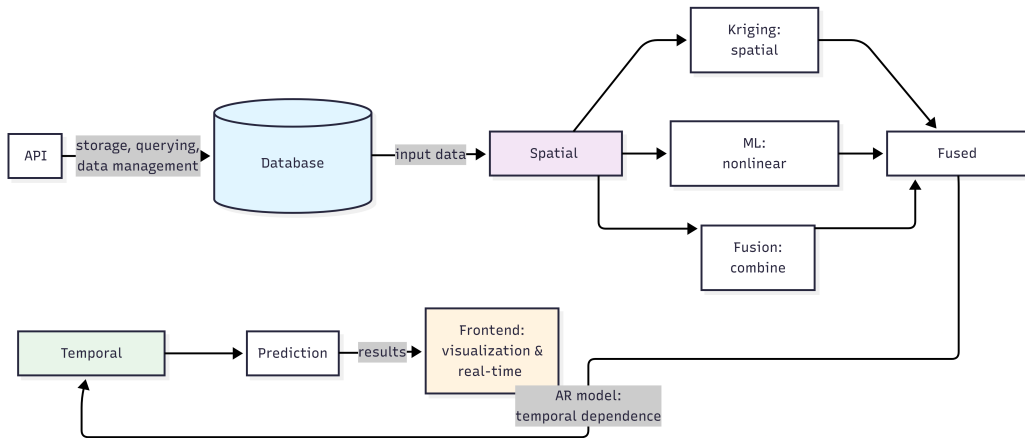
Motivation

- Weather station data exhibit:
 - Temporal dependence
 - Spatial dependence
 - Non-stationarity
- Time series models ignore spatial structure
- Kriging assumes linearity and stationarity
- Machine learning ignores spatial dependence

Goal

Develop an end-to-end spatio-temporal forecasting system

System Workflow



Step 1: Data Ingestion

- Source: Central Weather Administration API
- Frequency: hourly update
- Raw data stored in:

`observations_hourly`

- Station metadata stored in:

`stations`

Step 1: data resource

Swagger
1.6.0

Select a definition openAPI

中央氣象署開放資料平臺之資料擷取API 1.0.0 OAS 2.0

[Base URL: /api]

提供目前開放資料之擷取API

Schemes
HTTPS

觀測

GET /v1/rest/datastore/O-A0001-001 氣象觀測站-全測站即時氣象資料

氣象觀測站-全測站即時氣象資料

Try it out

Name	Description
Authorization required	氣象開放資料平台會員授權碼
string (query)	Authorization
limit	限制最多回傳的資料，預設為回傳全部筆數
number (int) (query)	limit
offset	指定從第幾筆後開始回傳，預設為第 0 筆開始回傳
number (int) (query)	offset
format	回傳資料格式，預設為 json 格式
string (query)	Available values: JSON, XML --
StationId	測站站號，請參考 https://hdp.cwb.gov.tw/static/state.html ，可輸入多個站號，大小寫必須完全符合，預設為全部回傳
array of strings (query)	
StationName	測站站名，請參考 https://hdp.cwb.gov.tw/static/state.html ，可輸入多個站名，預設為全部回傳，若使用「StationId」，則參數「StationName」的篩選資料將會失效，只會回傳StationId符合的資料
array of strings (query)	

Step 1:data of all station

	id	station_id	station_name	datetime	lat	lon	temperature	humidity	wind_speed	wind_direction	pressure
	822	CAP040	國一N361K	2026-04-13 10:00:00	22.680555	120.328611	29.6	72	1.8	200	1011.2
	823	CAP060	國三S385K	2026-04-13 10:00:00	22.767777	120.444444	28.8	62	1.2	80	1003.2
	824	CAP070	高雄燈塔	2026-04-13 10:00:00	22.614594	120.26468	27.5	75	1.8	230	1011.1
	825	CAQ010	國三N425K	2026-04-13 10:00:00	22.480277	120.531666	28.3	70	1.5	220	1010.5
	826	CAQ020	國三S415K	2026-04-13 10:00:00	22.568	120.528	28.7	64	1.3	280	1011
	827	CAQ030	崎峰國小	2026-04-13 10:00:00	22.42565	120.49521	NULL	NULL	2.6	240	1014.1
	828	CAUB30	國五S031K	2026-04-13 10:00:00	24.819144	121.782378	28	78	2.4	50	1012.1
	829	CAUB40	國五S041K	2026-04-13 10:00:00	24.734764	121.781922	28.1	76	1.3	90	1010.1
	830	CAUB50	國五S047K	2026-04-13 10:00:00	24.675456	121.794583	27.7	75	2.2	60	1010.1
	831	CAV270	國一S336K	2026-04-13 10:00:00	22.894167	120.271178	28.6	72	2.4	200	1010.9
	832	CAV280	國三S378K	2026-04-13 10:00:00	22.825958	120.415519	28.3	75	1.5	280	1001.2
	833	CAIW010	東引	2026-04-13 10:00:00	26.358836	120.483631	19.1	98	1.6	120	1007.8
	834	E2H360	鹽埕池	2026-04-13 10:00:00	23.918333	120.885278	27.3	53	1.9	130	940.7
	835	E2HA20	畢祿溪站	2026-04-13 10:00:00	24.226944	121.308522	20.5	17	0.7	60	790
	836	E2K600	四湖植物園	2026-04-13 10:00:00	23.676986	120.163433	29.3	86	1.6	240	1014.1
	837	E2P980	六龜林試	2026-04-13 10:00:00	22.991703	120.645722	28.8	53	0.9	180	984.5
	838	E2P990	扇平林試	2026-04-13 10:00:00	22.9677	120.685703	NULL	NULL	NULL	NULL	NULL
	839	E2S960	太麻里(2)	2026-04-13 10:00:00	22.598	120.99775	30.6	56	2.2	170	1012
	840	E2S980	太麻里(1)	2026-04-13 10:00:00	22.598922	120.980044	30.1	56	0.9	120	1003.7
	841	G2F820	農業試驗所	2026-04-13 10:00:00	24.031261	120.688133	28.9	67	3.1	260	1005.3
	842	G2L020	農試嘉義分所	2026-04-13 10:00:00	23.485028	120.473806	29.4	62	2.4	250	1005.9
	843	G2M350	農試溪口農場	2026-04-13 10:00:00	23.58565	120.408111	27.9	70	2.9	230	1012.3
	844	G2P820	農試鳳山分所	2026-04-13 10:00:00	22.646611	120.356083	29.9	57	1.1	150	1011.3
	845	K2E360	苗栗農改	2026-04-13 10:00:00	24.495722	120.829111	30.2	53	2.6	260	1000.6
	846	K2E710	苗改生物防...	2026-04-13 10:00:00	24.422861	120.871833	29.2	49	1.8	190	981.3
	847	K2F750	植苗繁殖	2026-04-13 10:00:00	24.225972	120.800917	26.2	62	2.3	240	961.8
	848	U2H480	臺大溪頭	2026-04-13 10:00:00	23.670306	120.798194	23.3	64	0.9	320	887.3
	849	U2HA30	臺大和社	2026-04-13 10:00:00	23.590856	120.888864	27.4	58	1	80	927.3
	850	U2HA40	臺大內茅埔	2026-04-13 10:00:00	23.689722	120.851117	29.1	45	0.6	30	956.8
	851	U2HA50	臺大竹山	2026-04-13 10:00:00	23.756972	120.681556	NULL	NULL	NULL	NULL	NULL
	852	V2K610	大庄合作社	2026-04-13 10:00:00	23.8068	120.367939	27.9	74	2.3	230	1012.4
	853	V2K620	麥寮合作社	2026-04-13 10:00:00	23.753836	120.294783	28.2	66	3.1	230	NULL

Step 1: data of one station in different time

	id	station_id	station_name	datetime	lat	lon	temperature	humidity	wind_speed	wind_direction	pressure
▶	1	C0TB40	崇德	2026-04-13 10:00:00	24.166144	121.657414	29.1	66	1.6	32	1011.3
	854	C0TB40	崇德	2026-04-13 11:00:00	24.166144	121.657414	30.3	64	1.4	47	1010.7
	1707	C0TB40	崇德	2026-04-13 12:00:00	24.166144	121.657414	29.2	67	2.1	74	1010.1
	2560	C0TB40	崇德	2026-04-13 13:00:00	24.166144	121.657414	28.1	73	2.4	115	1009.5
	3413	C0TB40	崇德	2026-04-13 14:00:00	24.166144	121.657414	27.9	70	3.4	134	1008.8
	4266	C0TB40	崇德	2026-04-13 15:00:00	24.166144	121.657414	27.6	72	2.2	128	1008.4
	5119	C0TB40	崇德	2026-04-13 16:00:00	24.166144	121.657414	27.4	72	2.3	130	1008.4
	5972	C0TB40	崇德	2026-04-13 17:00:00	24.166144	121.657414	27.6	72	1	155	1008.4
	6825	C0TB40	崇德	2026-04-13 18:00:00	24.166144	121.657414	26.6	75	0.7	272	1008.9
	7678	C0TB40	崇德	2026-04-13 19:00:00	24.166144	121.657414	26.1	76	0.6	261	1009.5
	8531	C0TB40	崇德	2026-04-13 20:00:00	24.166144	121.657414	25.5	78	1	248	1010.2
	9384	C0TB40	崇德	2026-04-13 21:00:00	24.166144	121.657414	24.6	80	1.2	242	1010.7
	10237	C0TB40	崇德	2026-04-13 22:00:00	24.166144	121.657414	24.4	79	0.9	289	1010.8

Step 2: Spatial Modeling

Kriging

- Uses spatial coordinates
- Based on variogram
- Provides spatially consistent prediction

Machine Learning

- Uses spatial coordinates (x, y)
- Learns nonlinear spatial relationships
- Implemented via ensemble (Super Learner)

Key Idea

Kriging models spatial structure, ML captures nonlinear spatial patterns

Step 3: Model Fusion

$$\hat{Z}_{fusion}(u, t) = w(u)\hat{Z}_{ML}(u, t) + (1 - w(u))\hat{Z}_{KR}(u, t)$$

Where

$$w(u) = (\sigma_{KR}^2)^b, \quad b = b_0 + b_1 Z(u)$$

- $\hat{Z}_{ML}(u, t)$: ML prediction
- $\hat{Z}_{KR}(u, t)$: Kriging prediction
- $w(u)$: weight based on kriging variance
- Because σ^2 is standardize ,the interval between $w(u)$ is $[0, 1]$

Step 3: Model Fusion

Property :

- If the value near the station, then Kriging model has the main influence.
- If the value far then station, then machine learning help us to predict the value.
- In complex situation like non-stational, trend, skew etc, mix model is better than only kriging.
- In simple situation like Gaussian, Kriging is little better than mix.

Step 4: Temporal Modeling (AR)

- Apply autoregressive (AR) model at each location

- Input:

$$\hat{Z}_{fusion}(x, y, t - 1), \hat{Z}_{fusion}(x, y, t - 2), \dots$$

- Output:

$$\hat{Z}_{final}(x, y, t)$$

Step 5: Prediction Storage

Results are stored in:

`predictions_hourly`

- Kriging prediction
- ML prediction
- Fusion prediction
- AR-based final prediction

Database Role

Data storage, prediction tracking, and frontend query support

Combined Machine Learning and Kriging Approach

- Reference:
 - Erten, Yavuz, and Deutsch (2023)
 - *Estimation with a Combined Machine Learning and Kriging Spatial Estimation*

Machine Learning

- Captures nonlinear relationships
- Handles non-Gaussian data

Kriging

- Models spatial dependence
- Provides unbiased interpolation

Temporal Model (AR)

- Captures temporal dependence
- Models time series dynamics

Expected Results

- Real-time spatio-temporal forecasting pipeline
- Comparison:
 - Kriging
 - Machine Learning
 - Combined spatial model
- Outputs:
 - Spatial heatmap
 - Temporal forecasts (time series)
 - Error analysis

Conclusion

- Spatial modeling is achieved via Kriging and ML
- Temporal dependence is modeled separately using AR
- The proposed system separates spatial and temporal structures
- Fusion model leverages strengths of both spatial and temporal modeling

To do list

- Compare with the weight where $w(u) = 0, 1/2, 1$
- Think that how can i design with the interval b_0, b_1
- Test which stardardize of the $\sigma_{KR}^2(u)$ can be use (reference doesn't talk to this)